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Tissue Pressure and Oxygenation Monitoring for Anterior Cervical Spinal Surgery

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Abstract

- Anterior cervical discectomy and fusion (ACDF) requires sustained tissue retraction, which can cause dysphagia and dysphonia, yet current retractors provide no feedback on applied force.
- We developed a sensing retractor integrating a load cell (force), pressure sensors (tissue stress), and SpO₂ sensors (tissue perfusion) to enable real-time intraoperative monitoring.

Introduction and problem statement

- Surgical access during ACDF requires sustained retraction of the trachea, esophagus, and surrounding soft tissues, placing delicate structures under continuous mechanical load.
- Excessive force and prolonged retraction have been strongly associated with postoperative complications such as dysphagia and dysphonia, affecting up to 79% of patients.
- Despite these known risks, current retractors are purely mechanical and provide no quantitative feedback on applied force, pressure distribution, or tissue perfusion.
- Problem:** There is no system that provides real-time, quantitative monitoring of tissue pressure and perfusion during ACDF, limiting data-driven intraoperative decision-making and increasing the risk of soft-tissue injury.

Design specifications

Criteria

Specification

Compatible with existing retractor models

Removeable sensor attachments add < 0.25 lb.

Sensitive to physiologically relevant forces

Sensors have linear output in the 0 – 10 N range

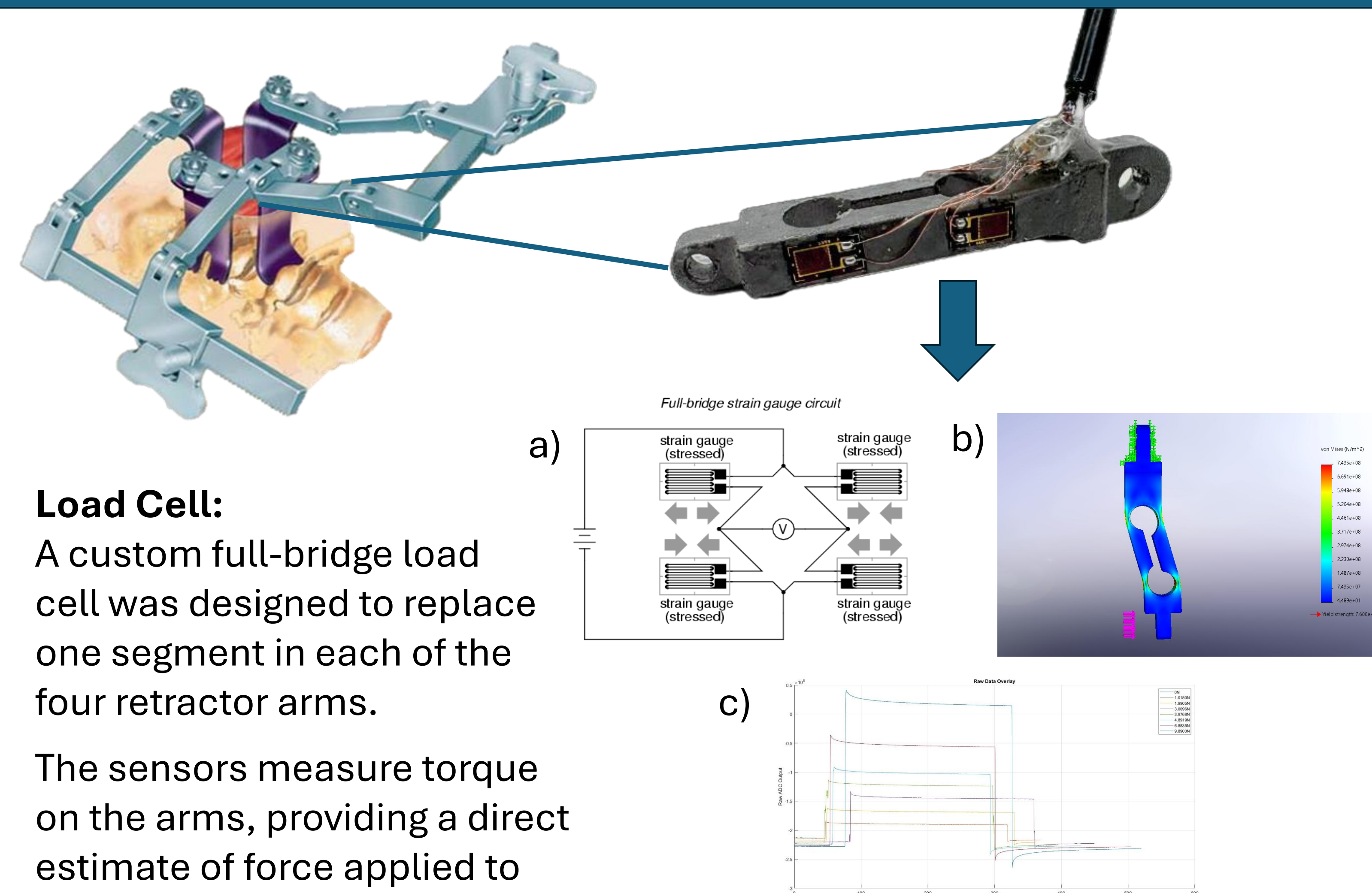
Real-time feedback

Continuous, low latency (<100-200 ms) sensor updates

Battery life

System runs on single battery for >4 hours

Retractor arm modifications



- Load Cell:** A custom full-bridge load cell was designed to replace one segment in each of the four retractor arms.
- The sensors measure torque on the arms, providing a direct estimate of force applied to surrounding tissue during surgery, in real time.

Figure 1 (a) Full-bridge circuit, (b) FEA-guided design, and (c) raw ADC output under applied loads.

Retractor blade modifications

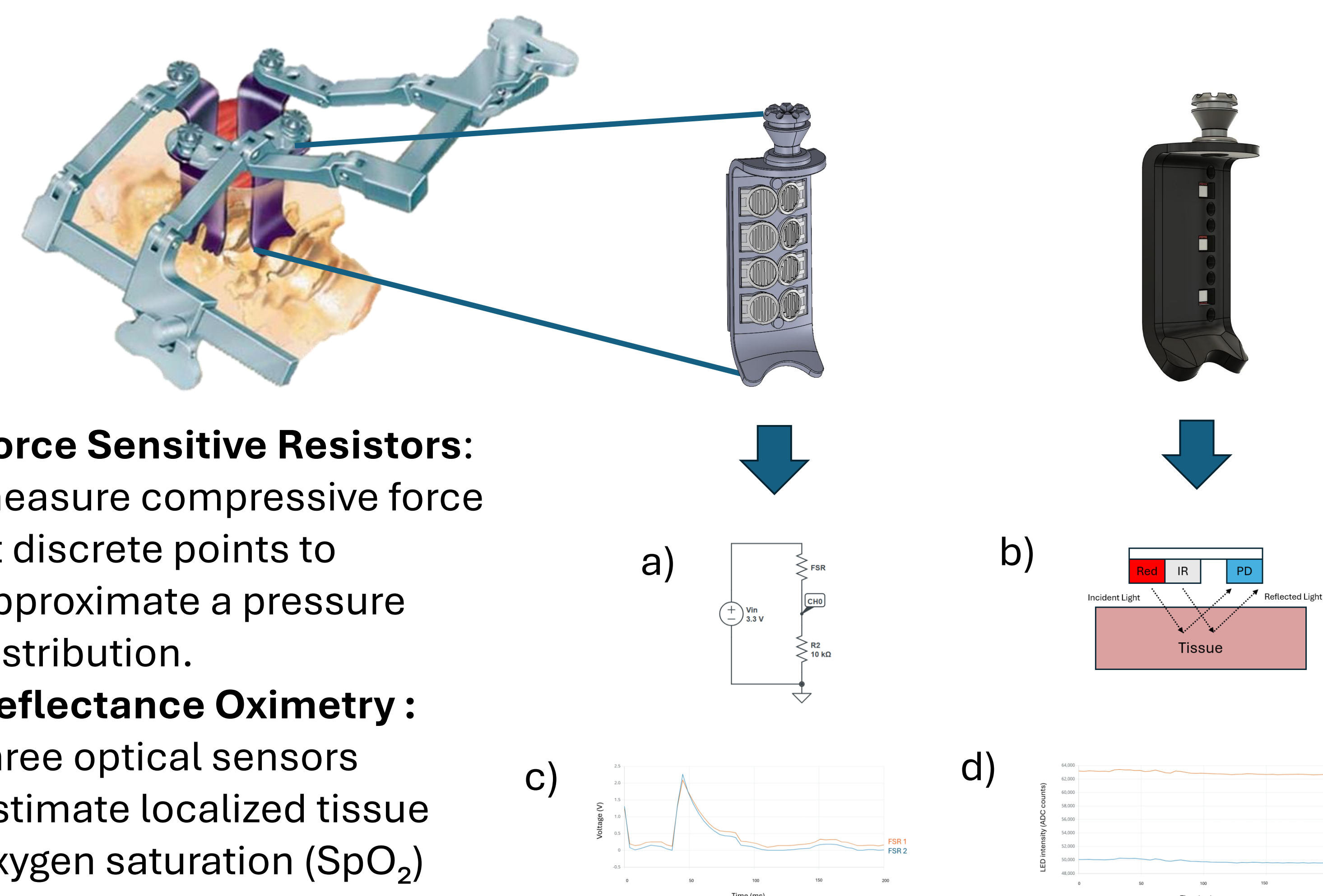


Figure 2 (a) Oximetry sensor diagram, (b) reflectance pulse oximetry sensing principle, (c) raw output from two FSRs, and (d) raw output from the oximetry sensor.

Sensor integration

- A dedicated multiplexer routes and processes signals from the pressure sensors, while a secondary I²C multiplexer manages multiple SpO₂ sensing channels.
- Sensor signals are acquired through a microcontroller and transmitted to an external computer for real-time processing and visualization.
- Custom-designed housing and mounting ensured stable sensor placement while minimizing interference with surgical workflow.

Testing

- Testing in this study focused on validation of individual sensing components, including force, pressure, and SpO₂ sensors, to ensure signal reliability and feasibility of integration within the retractor system.
- Component-level testing was performed to assess sensitivity, response consistency, and basic functionality under controlled conditions.

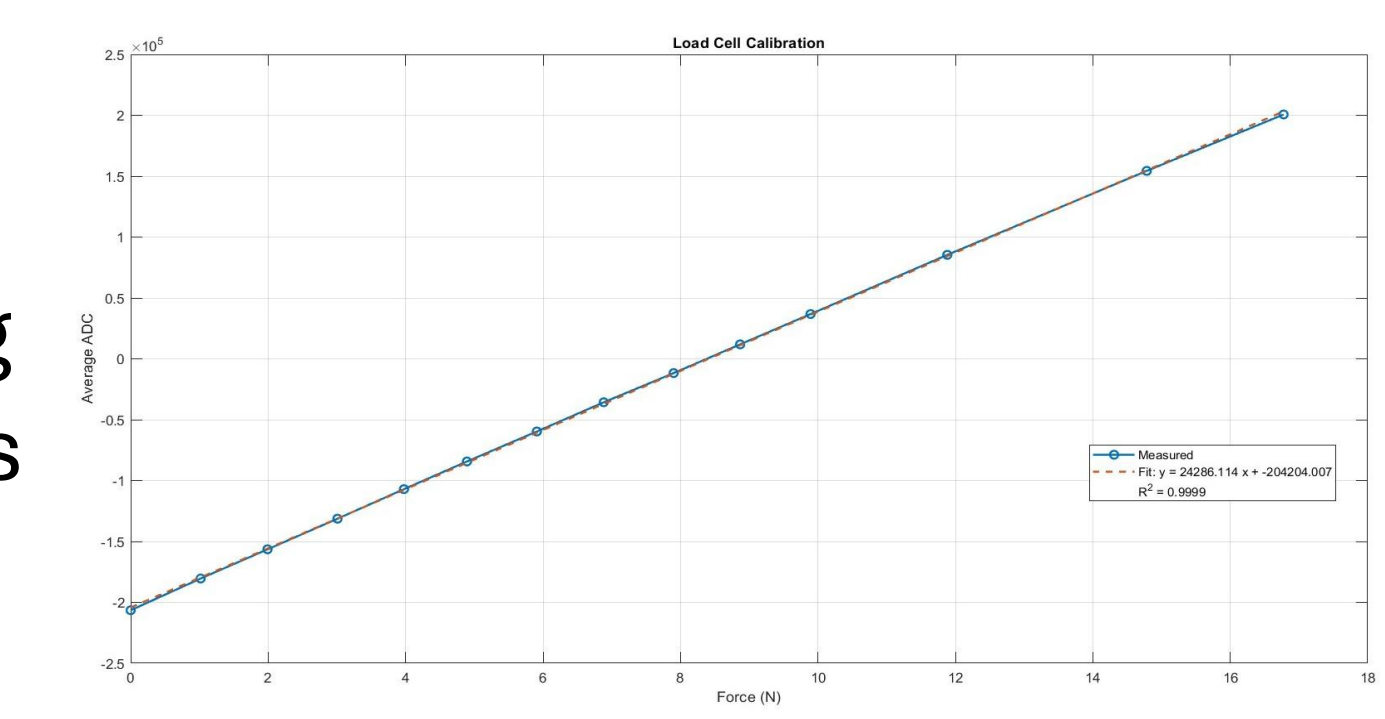


Figure 3: Linear calibration of ADC output vs. applied force ($R^2 = 0.9999$) for the load cell.

Conclusions and future work

- Our multimodal sensing retractor monitors force, pressure, and tissue perfusion in real time during ACDF surgery.
- The design preserves surgical workflow while potentially enabling data-driven decision-making to reduce soft-tissue injury and improve intraoperative consistency.
- Future work focuses on sensor validation through bench, cadaveric, and clinical testing.

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